



Multimodal Stance Detection Integrating Text and Visual Evidence in AI Argument Mining

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Abstract

Persuasive communication on social media is nowadays multimodal, combining textual and visual elements to express and frame argumentative positions. While transformer language models achieve strong performance in text-only stance detection, it remains unclear to what extent visual information provides complementary evidence beyond these strong textual baselines.

In this work, we investigate multimodal fusion for binary stance detection on the ImageArg Shared Task 2023 dataset, which consists of tweet-image pairs on controversial topics such as gun control and abortion. We first establish strong textual baselines using transformer models, where DeBERTa-v3-base achieves an F1-score of 0.835 (positive class = support) with synonym-based data augmentation. We then evaluate multiple vision backbones (ResNet50, ViT, CLIP, BLIP, ALIGN) in a unimodal setting, where CLIP obtains the highest visual-only performance ($F1 = 0.628$), indicating that images contain stance-relevant cues.

Building upon these baselines, we design multimodal architectures combining DeBERTa-v3 with pretrained vision encoders through early fusion strategies (mean pooling, weighted mean, gated fusion, concatenation, projected concatenation). The best multimodal configuration (DeBERTa + ALIGN with projected concatenation and caption augmentation) achieves an F1-score of 0.861, corresponding to an improvement of approximately +2.5 F1 points over the strongest text-only baseline.

Our findings show that language carries the dominant argumentative signal; however, carefully designed multimodal fusion consistently yields measurable gains. The effectiveness of multimodality strongly depends on fusion design and modality balancing. Overall, visual cues provide complementary information for stance detection, but only when integration strategies are thoughtfully orchestrated.

1 Introduction

Persuasive communication on social media has become increasingly multimodal. Political debates, advocacy campaigns, and controversial discussions are rarely expressed through text alone; instead, they frequently combine linguistic messages with images, memes, short videos and symbolic visual content. In such contexts, visual material may reinforce, frame, or subtly modify the argumentative intent of the accompanying text.

Stance detection, the task of identifying whether a piece of content supports or opposes a given topic, is particularly challenging in this setting. Arguments are often indirect, emotionally framed, or you need context to understand them. The same textual statement, when paired with different images, can convey distinct or even opposing stances (See Figure 1). Therefore, meaning does not reside solely in the textual signal, but may emerge from the interaction between textual and visual cues.



Figure 1: Same textual statement paired with different images can imply different stances.

Despite this multimodal online persuasion, most state-of-the-art stance detection systems remain unimodal, primarily text-based. Transformer-based language models have achieved strong performance on stance classification tasks, suggesting that linguistic information carries a substantial portion of the explicit argumentative signal. However, it remains unclear to what extent visual information provides complementary evidence beyond these already strong textual baselines.

From a cognitive perspective, this question is particularly relevant. Human perception and reasoning are multimodal. Neuroscientific research suggests that the brain integrates linguistic and visual information when constructing meaning and evaluating persuasive content (Vigliocco et al., 2014). Inspired by this integrative processing, multimodal AI systems attempt to combine heterogeneous representations to improve semantic understanding and decision-making.

In this work, we investigate whether multimodal models can improve stance detection in argumentative social media content. Using the ImageArg dataset, we systematically compare strong text-only baselines with multimodal architectures that integrate textual and visual representations through different fusion strategies. Our goal is not only to measure performance gains, but also to better understand how modality interaction affects stance prediction.

By analyzing unimodal and multimodal models under a unified experimental framework, this project aims to clarify when and how visual information contributes to stance detection, and whether carefully designed multimodal fusion can provide consistent improvements over strong textual baselines.

2 Related Work

2.1 Natural Language Processing

In the past few years, Natural Language Processing (NLP) has experienced considerable progress, primarily due to the development of large language models (LLMs) built on transformer architectures (Vaswani et al., 2017). Unlike recurrent or convolutional models, transformers rely on self-attention mechanisms that allow each token in a sequence to attend to all other tokens simultaneously. This architecture enables the modeling of long-range dependencies and contextual relationships more effectively than previous sequential approaches. Large Language Models (LLMs) built upon the transformer encoder architecture learn contextualized token representations through multi-head self-attention and feed-forward layers. These contextual embeddings can capture semantic, syntactic, and discourse-level information. These models have achieved state-of-the-art performance on diverse NLP tasks, such as text classification, sentiment analysis, and stance detection.

2.2 Argument Mining

Argument Mining (AM) seeks to automatically identify argumentative structures and their relations within natural language (Cabrio and Villata, 2018; Lawrence and Reed, 2020). AM aims to detect the components of an argument and model how they interact.

An argument is typically composed of *claims* and *premises*. A *claim* represents the main standpoint or conclusion, while a *premise* provides justification or evidence supporting that claim. Beyond this, AM systems attempt to detect the relations between them. A *support* relation indicates that a premise strengthens or justifies a claim, whereas an *attack* relation signals contradiction or opposition between argumentative components.

Early research in AM focused on identifying argumentative units, classifying argument com-

ponents (e.g., claim vs. premise), and detecting support and attack relations within structured texts. A subsequent line of work extended these methods to political debates and social media, where systems aim to determine how speakers construct arguments, defend positions, or oppose specific viewpoints (Menini and Others, 2018; Haddadan et al., 2019). In this context, stance detection can be seen as a related task, focusing on identifying whether a given text expresses support or opposition toward a predefined target.

More recent studies broaden the scope of AM to include the detection of fallacies and cognitive biases in argumentative discourse (Goffredo and Others, 2022, 2023; Mou and Others, 2023; Agiza et al., 2024). A *fallacy* refers to a flawed pattern of reasoning that may appear persuasive but lacks logical validity (e.g., ad hominem attacks or appeals to emotion). Detecting such phenomena illustrates the complexity of modeling argumentative language computationally.

2.3 Multimodal NLP and Argument Mining

While traditional argument mining focused primarily on textual analysis, there is a growing interest in *multimodal argument mining*, where visual information complements textual data. Multimodal fusion strategies, such as early, intermediate, and late fusion, have been explored to combine visual and textual features effectively (Boulahia and Others, 2021; Shin et al., 2022). Additionally, recent advances in Large Vision-Language Models (VLMs) have enabled the learning of shared cross-modal representations that integrate linguistic and visual signals.

In this context, two related tasks are particularly relevant. *Argument detection* aims to identify whether a given text (or multimodal instance) contains an argument and to extract its argumentative components. *Stance classification* focuses on determining whether the content expresses support or opposition toward a predefined target or topic.

Multimodal datasets such as ImageArg (Liu et al., 2023) are composed of text-image pairs annotated for stance and persuasiveness. Other related datasets include M-Arg (Mestre et al., 2021) and MM-Claims (Cheema and Others, 2022). These resources enable research on how linguistic and visual information interact to improve both argument detection and stance classification in socially relevant domains.

2.4 Neuroinspired Perspective

From a cognitive perspective, multimodal argument mining is similar to the way human beings combine evidence across sensory modalities. Neuroscience and cognitive science have shown that at least in processing language, the brain combines linguistic and visual information to create coherent judgments and evaluate persuasion (Vigliocco et al., 2014). This analogy inspired the design of multimodal AI agents, which can imitate aspects of human reasoning to make sense

of arguments presented in language and in images.

3 Research Question and Objectives

The rapid progress of LLM has significantly improved performance on stance detection tasks in purely textual models. However, in social media context, arguments are frequently expressed through a combination of text and images. Although multimodal approaches have demonstrated strong results in several classification tasks, it remains unclear to what extent visual information provides complementary evidence beyond strong text-only baselines in controversial debates.

The present work investigates the following research question:

Does multimodal fusion improve binary stance detection performance on the ImageArg dataset?

From this research question, we derive two main hypotheses:

- H1. Multimodal fusion models outperform unimodal text-only models evaluated with Binary F1-Score (Positive class = support).
- H2. Visual representations provide complementary information that improves stance prediction beyond textual signals alone.

The evaluation follows the official ImageArg Shared Task protocol, focusing on the Binary F1-Score for the positive class (*support*). This choice reflects the importance of correctly identifying supportive stances in highly polarized debates. This metric is informative under class imbalance, as it balances precision and recall for the positive class.

Beyond testing these hypotheses, the broader objective of this internship is experimental: To benchmark unimodal and multimodal architectures on the ImageArg dataset, explore different fusion strategies, and iteratively improve performance toward state of the art levels reported in recent multimodal stance detection studies.

4 Dataset

The **ImageArg Shared Task 2023** dataset (Liu et al., 2023) is a multimodal collection of tweets with images, designed to explore the tasks of stance and persuasiveness classification. The dataset focuses on two highly debated social topics: *gun control* and *abortion*. Each

observation consists of a tweet text, an associated image, and annotations indicating the stance of the argument (*support* or *oppose*), and the persuasiveness of the image (*yes* or *no*).

The dataset contains 2,314 observations, divided into: 1,814 for training, 200 for validation, and 300 for testing. Since each tweet is paired with a corresponding image, this ensures that both textual and visual modalities are available for multimodal modeling.

Examples

To illustrate the multimodal nature of the dataset, Figure 2 presents representative examples across both stance labels and topics.

In the abortion related example (top row), visual content has explicit textual advocacy messages (e.g., campaign slogans) and emotional framing (e.g., baby’s feet), which may reinforce or frame the textual argument. In the gun control examples (bottom row), some images present symbolic political graphics, while others display emotionally charged photographs. In several cases, images contain embedded textual elements (e.g., slogans or campaign material), suggesting that visual models may capture stance-relevant cues both through visual symbolism and written content within images.

These examples highlight how visual content can amplify, contextualize, or emotionally frame the textual message, motivating the exploration of multimodal modeling strategies.



Figure 2: Representative examples from the ImageArg dataset. Top: abortion-related tweets labeled as Support and Oppose. Bottom: gun control-related tweets labeled as Support and Oppose. Each instance consists of a tweet and an associated image, illustrating the multimodal argumentative structure of the dataset.

The distribution of stances in the training set is imbalanced, with 1,095 *oppose* instances and 719 *support* instances. This class imbalance is an important consideration for model training and evaluation.

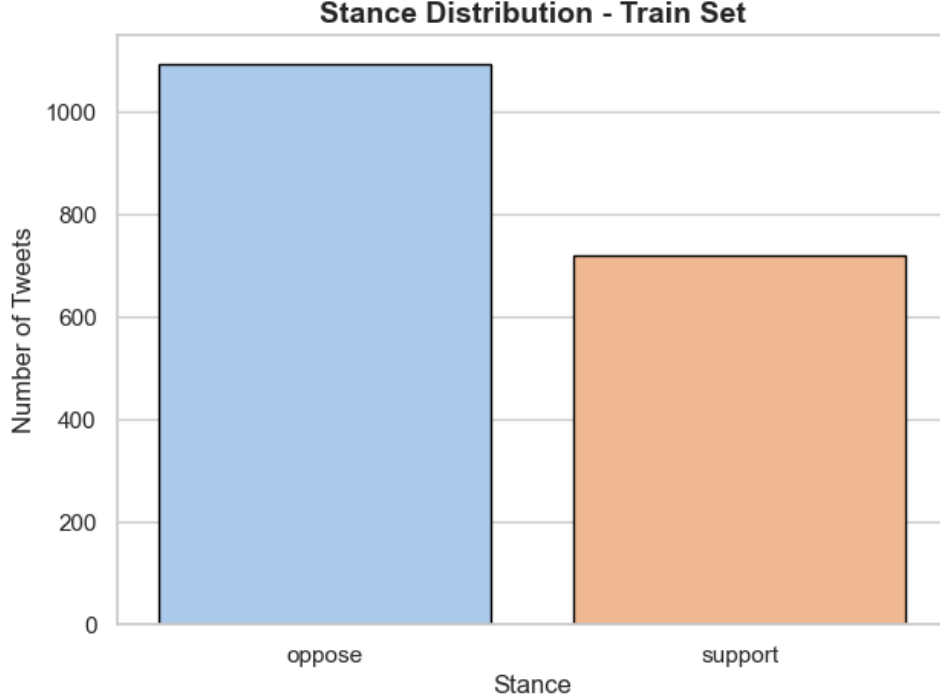


Figure 3: Distribution of argumentative stances in the training split of the ImageArg Shared Task 2023 dataset.

No manual text normalization was applied prior to tokenization; tweets may contain URLs, hashtags, mentions, and other typical social media artifacts. Images were processed using the default preprocessing pipelines associated with each pretrained vision backbone.

Overall, the ImageArg dataset provides a challenging benchmark for multimodal stance detection, as argumentative signals may appear explicitly in text, implicitly through visual framing, or through the interaction between both modalities.

5 Methodology

5.1 Dataset and Data Splits

All experiments were conducted using the ImageArg dataset. The original train, dev, and test partitions were preserved without modification to ensure comparability with prior work.

Hyperparameter tuning and model selection were based on performance on the development set. The test set was used for final evaluation.

The original training set exhibited moderate class imbalance. To address this issue, we applied data augmentation techniques exclusively to the training split, resulting in a balanced distribution:

- Train (after augmentation): 1095 Oppose, 1095 Support
- Dev: 127 Oppose, 73 Support
- Test: 182 Oppose, 118 Support

5.2 Data Augmentation

To address class imbalance in the training split, several textual data augmentation strategies were evaluated on the minority class (*support*). The goal was to generate additional synthetic samples to match the number of *oppose* instances.

The following augmentation techniques were tested:

- Synonym replacement
- Back-translation
- Paraphrasing
- Noise injection

Each technique was evaluated by training DeBERTa-v3-base and selecting the best augmentation based on development-set Binary F1 (support). The final comparison is reported on the test set for transparency. Among the tested strategies, synonym replacement yielded the highest F1-Score on the positive class. A total of 376 synthetic samples were generated to balance the minority class, resulting in an equal class distribution in the augmented training set.

The augmented dataset was generated once and saved, ensuring that all subsequent experiments were conducted on a fixed and reproducible dataset.

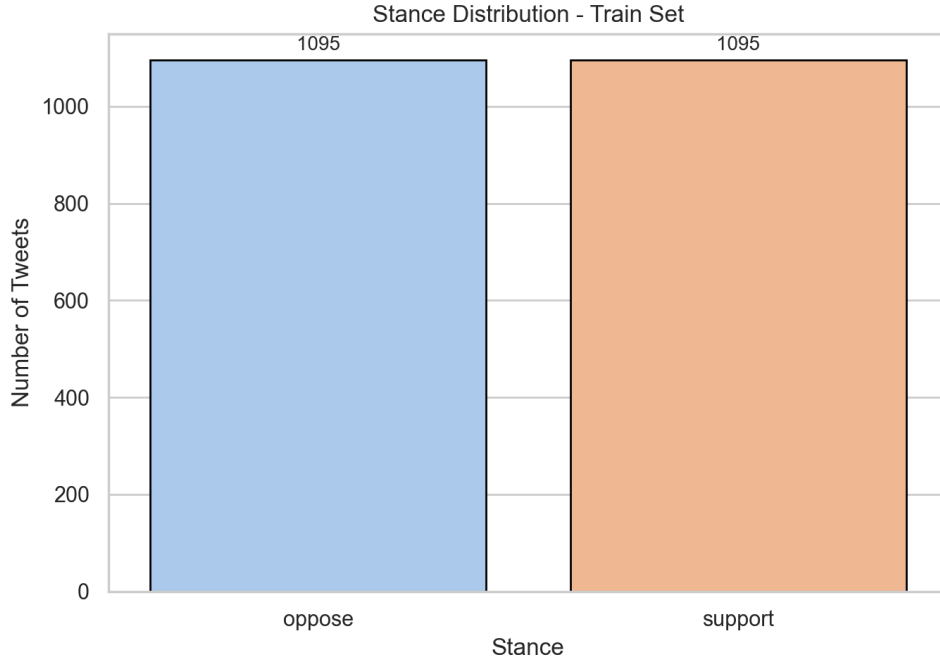


Figure 4: Distribution of argumentative stances in the training split of the ImageArg Shared Task 2023 dataset after augmentation.

5.3 Textual Models

To establish a textual baseline for stance detection, we evaluated both traditional machine learning models and transformer-based architectures.

Classical baselines. As a reference point, we implemented Logistic Regression (LR) and Support Vector Machines (SVM) using TF-IDF representations of the tweets. These models provide a classical comparison framework commonly used in early stance detection research, allowing us to quantify the performance gap between traditional approaches and modern pre-trained language models.

Transformer-based models. We evaluated several transformer-based encoders that have demonstrated strong performance on social media and stance-related tasks:

- bert-base-uncased
- roberta-base
- cardiffnlp/twitter-roberta-base
- vinai/bertweet-base

- microsoft/deberta-base
- microsoft/deberta-v3-base

These models were selected due to their state-of-the-art performance in tweet classification, sentiment analysis, and stance detection tasks reported in recent literature. In particular, some models such as BERTweet and Twitter-RoBERTa are specifically pretrained on social media corpora, while DeBERTa-v3 has demonstrated strong generalization capabilities across multiple NLP benchmarks.

Each model was fine-tuned for binary classification with two output labels (*oppose*=0, *support*=1). Text inputs were tokenized using the corresponding pretrained tokenizer, with truncation and padding to a fixed maximum length of 128 tokens.

All transformer models were trained on the synonym-augmented training dataset. Training was conducted using Hugging Face. The model architecture is composed by the pre-trained model and a classification head on top (dropout layer followed by a linear classifier applied to the transformer representation).

Models were optimized using AdamW with a learning rate of 2×10^{-5} , weight decay of 0.01, batch size of 16, and up to 10 epochs. Evaluation on the development set was performed at the end of each epoch. The best checkpoint was selected according to the Binary F1-Score (positive class = *support*) on the development set. Early stopping was applied with a patience of 3 epochs to reduce overfitting.

Experiments were conducted using PyTorch 2.9.1 and Transformers 4.57.0 on an NVIDIA H100 NVL GPU.

5.4 Vision Models

After establishing strong text-only baselines, we evaluated vision models to determine whether images, by themselves, contain stance cues that could complement language. This unimodal vision step serves two purposes:

1. To quantify how informative the visual modality is in isolation
2. To select robust visual encoder capable of extracting meaningful representations for later the multimodal fusion.

We followed a progressive modeling strategy, we started with a classical convolutional network baseline and then moving to modern transformer vision encoders. This enables a clearer interpretation of the performance gap between traditional computer vision approaches and large pre-trained models.

All vision models were trained for binary stance classification (*oppose*=0, *support*=1). Model selection was based on the performance, with the Binary F1-Score for the positive class (*support*) as the primary metric.

ResNet50. As a classical convolutional computer vision baseline, we implemented ResNet50 pretrained on ImageNet. Images were resized to 224×224 and normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). The final fully connected layer (2048 → 2) was replaced to perform binary classification.

To study the trade-off between generalization and adaptation on a relatively small dataset, we tested three freezing strategies: (i) fully frozen encoder, (ii) partial fine-tuning (training only the last convolutional block, layer4), and (iii) full fine-tuning.

Training hyperparameters were: batch size 32, learning rate 2×10^{-5} , weight decay 1×10^{-4} , maximum epochs 10, and early stopping patience of 3 epochs.

CLIP (vision encoder). We evaluated the vision component of `openai/clip-vit-base-patch32`. CLIP was selected because it is pretrained with image-text contrastive learning, making its visual representations potentially more aligned with semantic concepts expressed in social media text. The CLIP vision transformer produces 768-dimensional hidden representations, which are projected into a 512-dimensional embedding space via the pretrained visual projection layer.

For stance classification, the CLS token from the last hidden state was extracted, projected to 512 dimensions, and passed through a trainable classification head: dropout (0.3) → linear (512 → 256) → ReLU → dropout (0.2) → linear (256 → 2).

We explored freezing strategies analogous to ResNet50, including partial fine-tuning of the last transformer layer.

Training hyperparameters were: batch size 16, learning rate 2×10^{-5} , weight decay 1×10^{-4} , warmup ratio 0.1, maximum epochs 10, and early stopping patience of 3 epochs.

Vision Transformer (ViT). We evaluated `google/vit-base-patch16-224` pretrained on ImageNet-21k as a strong vision-only transformer baseline. The original classification head was replaced to perform binary classification, and both partial and full fine-tuning strategies were tested. The model outputs logits directly from the CLS token representation.

Training hyperparameters were identical to those used for CLIP-based experiments.

BLIP vision encoder. We also evaluated the vision component of Salesforce/blip-base. BLIP is pretrained on large-scale vision-language data and has demonstrated strong multimodal understanding capabilities. In the unimodal setting, the vision encoder outputs pooled visual representations, which were passed through a linear classification head to predict stance labels. Frozen and fine-tuned variants were considered.

ALIGN vision encoder. Finally, we considered the vision encoder of kakaobrain/align-base. ALIGN was selected due to its large-scale vision-language pretraining, which can lead to semantically richer visual representations than those learned purely from ImageNet supervision. In the unimodal setting, ALIGN produces 640-dimensional visual embeddings, which were passed through a linear classification head to predict stance labels.

Both frozen and fine-tuned configurations were considered. When required for architectural consistency with later multimodal experiments, embeddings were projected to a common dimensionality (768).

For CLIP, ViT, BLIP, and ALIGN, we used the default preprocessing pipelines provided by their respective processors to ensure compatibility with pretrained weights. This includes resizing, normalization, and patch/token preparation consistent with the original pretraining setup.

In general, this vision stage provides a controlled assessment of how much stance information is present in images alone, and it motivates the need for multimodal fusion when visual evidence is not enough in isolation but still may be complementary to text.

5.5 Multimodal Architectures

After we identified strong unimodal baselines, we moved to multimodal modeling to combine linguistic and visual evidence for stance detection. Our multimodal pipeline encodes each modality independently and integrates them using fusion strategies designed to capture complementary information.

5.5.1 Overall Pipeline

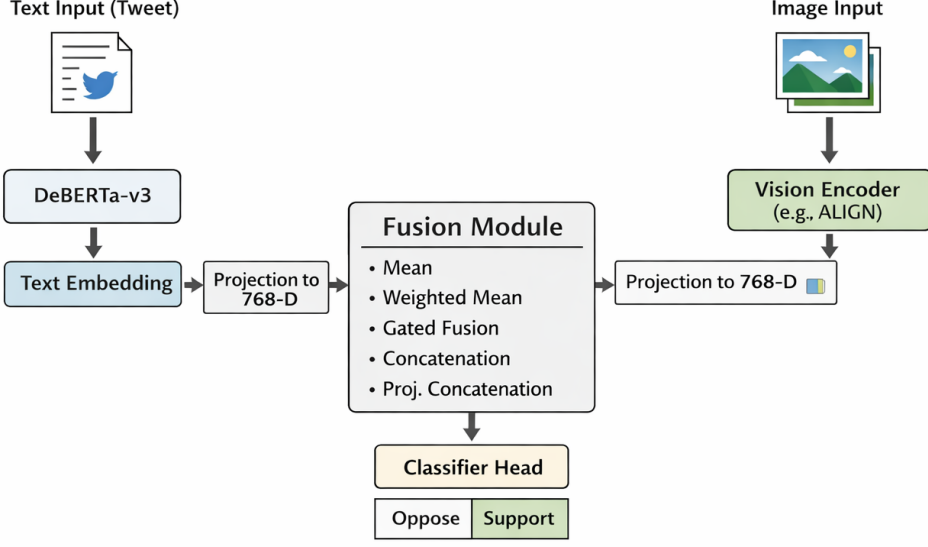


Figure 5: **Overview of the multimodal stance detection pipeline.** The tweet text is encoded with DeBERTa-v3 to obtain a textual embedding, while the image is encoded with a vision backbone (e.g., ALIGN) to obtain a visual embedding. Both embeddings are projected to a shared 768-dimensional space and combined using a fusion operator (mean, weighted mean, gated fusion, concatenation, or projected concatenation). The fused representation is then passed to a classifier head to predict stance (*oppose* vs. *support*).

All multimodal configurations used `microsoft/deberta-v3-base` as the text encoder, selected as the strongest text-only model in preliminary experiments. For the visual modality, we evaluated several pretrained vision and vision-language encoders (focusing mainly in our strongest ones: CLIP, BLIP, and ALIGN). While unimodal vision performance provides a reference, the main selection criterion for the multimodal stage was the effectiveness of the visual encoder as a feature extractor fused with the text.

Given that text and vision encoders may produce embeddings of different dimensionalities, we mapped both modalities into a shared representation space of dimension 768 (`COMMON_DIM=768`). This choice matches the hidden size of DeBERTa-v3 and simplifies fusion operations. When an encoder’s output dimension differed from 768, a linear projection layer was applied:

- Text projection: $\mathbb{R}^{d_t} \rightarrow \mathbb{R}^{768}$
- Vision projection: $\mathbb{R}^{d_v} \rightarrow \mathbb{R}^{768}$

Text embeddings were extracted from the [CLS] token of the final hidden state. Visual embeddings were obtained from the pooled output of the vision encoder (or its equivalent pooled representation). The fused representation was then passed to a classifier head.

5.5.2 Fusion Strategies

We primarily explored early fusion, where text and image representations are combined at the feature level before classification. We evaluated multiple fusion operators:

- **Mean pooling:** $h = \frac{1}{2}(t + v)$
- **Weighted mean:** $h = \alpha t + (1 - \alpha)v$, where α is a learnable parameter (normalized via sigmoid)
- **Gated fusion:** $g = \sigma(W[t; v])$, $h = g \odot t + (1 - g) \odot v$
- **Concatenation:** $h = [t; v]$
- **Projected concatenation:** $t' = P_t(t)$, $v' = P_v(v)$, $h = [t'; v']$

5.5.3 Late Fusion (Preliminary)

We conducted preliminary late-fusion experiments by combining modality-specific prediction scores; however, late fusion did not outperform early fusion and is therefore not detailed further.

5.5.4 Caption-Based Text Augmentation

To incorporate higher-level visual semantics in a text-dominant framework, we conducted caption-based experiments. Image captions were generated using pretrained captioning models (e.g., BLIP and ALIGN-based captioning), producing additional textual fields (e.g., `caption_blip`, `caption_align`). Captions were appended to the original tweet text and provided as augmented textual input to the DeBERTa encoder. This setting can be seen as a hybrid approach that injects visual information through language, enabling comparison with feature-level multimodal fusion.

5.5.5 Training Protocol

During multimodal training, the vision encoder was kept frozen, while the DeBERTa text encoder was fine-tuned end-to-end. This choice was because the small dataset size and the dominant role of textual information in stance detection.

Multimodal models were trained using AdamW with a learning rate of 2×10^{-5} , weight decay of 1×10^{-4} , batch size 16, and a linear learning-rate scheduler with warmup (warmup ratio 0.1). Training used cross-entropy loss and gradient clipping with a maximum norm of 1.0.

Model selection relied on early stopping based on development-set Binary F1-Score for the positive class (*support*). The best checkpoint (highest dev F1) was retained for final test evaluation.

6 Experiments and Results

6.1 Textual Results

6.1.1 Classical Baselines

We first evaluated traditional machine learning models using TF-IDF representations. Logistic Regression (LR) and Support Vector Machines (SVM) achieved very similar performance, with F1-Scores of 0.770 and 0.773 respectively (positive class = support). These results establish a classical baseline for comparison.

Table 1: Performance of classical text classifiers (test set).

Model	Accuracy	Precision	Recall	F1
Logistic Regression	79.7%	69.4%	86.4%	77.0%
SVM	79.7%	68.9%	88.1%	77.3%

6.1.2 Transformer-Based Models (Raw Training Data)

Transformer models substantially outperformed classical baselines. Table 2 reports mean and standard deviation across three random seeds.

Table 2: Performance of transformer-based models without data augmentation (mean $\pm$ std over 3 seeds).

Model	F1 (Mean)	F1 (Std)
DeBERTa-v3-base	0.827	0.009
RoBERTa-base	0.816	0.017
Twitter-RoBERTa	0.808	0.015
DeBERTa-base	0.788	0.014
BERTweet-base	0.787	0.006
BERT-base-uncased	0.765	0.023

Among all evaluated models, `microsoft/deberta-v3-base` achieved the highest mean F1-Score (0.8272) with relatively low variance across seeds, therefore was therefore selected as the textual model for subsequent multimodal experiments.

6.1.3 Impact of Data Augmentation

We evaluated multiple data augmentation strategies using DeBERTa-v3-base as the base model. Table 3 reports test performance for each augmentation method.

Table 3: Impact of data augmentation techniques on DeBERTa-v3-base (test set).

Technique	Accuracy	Precision	Recall	F1
Raw	0.833	0.716	0.940	0.813
Synonym	0.857	0.762	0.924	0.835
Back-translation	0.853	0.753	0.932	0.833
Paraphrase	0.843	0.742	0.924	0.822
Noise injection	0.790	0.669	0.924	0.776

Synonym augmentation yielded the highest F1-Score (0.8352), improving performance by +2.2 F1 points compared to the raw training setting. This suggests that moderate lexical variation in the minority class (*support*) improves generalization without significantly distorting semantic meaning.

Based on these results, synonym augmentation was adopted for subsequent textual and multi-modal experiments.

6.2 Vision-Only Results

To measure the independent contribution of visual information, we evaluated several vision backbones under a unimodal setting.

Table 4 reports test-set performance.

Table 4: Performance of vision-only models (test set).

Model	Accuracy	Precision	Recall	F1
ResNet50	0.587	0.475	0.475	0.475
CLIP	0.70	0.613	0.644	0.628
ALIGN	0.65	0.548	0.636	0.588
BLIP	0.693	0.628	0.542	0.582
ViT	0.587	0.474	0.458	0.466

Among all evaluated models, CLIP achieved the highest F1-Score (0.628), followed by ALIGN and BLIP. However, overall performance remains substantially lower than text-only transformers, which achieve F1-Scores above 0.82.

Although vision-only models underperform compared to text-based transformers, CLIP achieves an F1-Score of 0.6281, which is substantially above random performance. This indicates that visual encoders capture stance patterns to a certain extent. While images alone are insufficient for reliable stance classification, their performance suggests the presence of complementary information that may improve prediction when integrated with textual signals.

6.3 Multimodal Results

6.3.1 Early Fusion without Captions

We first evaluated early fusion strategies combining DeBERTa-v3-base with different visual encoders (CLIP, ALIGN, BLIP), without caption augmentation. Table 5 reports the best configurations per vision backbone.

Table 5: Best early fusion configurations without caption augmentation (test set).

Model	Fusion	F1	Accuracy
DeBERTa + CLIP	mean	0.8429	0.8633
DeBERTa + ALIGN	proj_concat	0.8516	0.8733
DeBERTa + BLIP	proj_concat	0.8506	0.8700

Among non-caption models, DeBERTa + ALIGN with projected concatenation achieved the highest F1-Score (0.8516), followed closely by DeBERTa + BLIP. All three vision encoders yielded improvements over the best text-only baseline ($F1 = 0.8352$), suggesting that visual information provides complementary cues when integrated at the feature level.

Late fusion did not outperform early fusion in preliminary experiments and therefore is not reported in detail

6.3.2 Caption-Augmented Experiments

We further then explored caption augmentation by appending generated image captions (BLIP2 or GIT) to the textual input and combining this approach with early fusion.

The best-performing configuration overall was:

- **DeBERTa + ALIGN + Caption (BLIP2) + projected concatenation:** $F1 = 0.8605$

Several caption-augmented configurations achieved F1-Scores above 0.85, outperforming non-caption variants. This suggests that incorporating visual semantics into the textual input can improve multimodal integration.

6.3.3 Comparison with Text-Only Baseline

The strongest text-only model (DeBERTa-v3-base with synonym augmentation) achieved $F1 = 0.8352$. The best multimodal configuration improved performance to 0.8605, corresponding to a gain of +2.5 F1 points.

While the improvements are moderate, they are observed across multiple fusion strategies and visual encoders. It is worth noting that not all multimodal configurations outperform the text-only baseline, suggesting that multimodal integration must be carefully designed to avoid introducing noise from the visual modality.

6.3.4 Ablation Analysis

A simple ablation experiment was conducted by selectively deactivating one modality during inference. Results indicate that textual representations contribute the majority of predictive power, while the visual modality provides a smaller but still measurable contribution. This confirms that stance detection in this dataset is primarily text-driven; however, visual cues can improve performance when carefully fused.

7 Discussion

7.1 Key Findings

From our experiments, key findings emerged.

First, strong textual baselines remain highly competitive in stance detection. Our transformer language models, particularly DeBERTa-v3-base, achieved high F1-Scores even without visual input. This confirms that language carries the dominant argumentative signal in the ImageArg dataset.

Second, multimodal models provide consistent and measurable improvements over text baselines. Although the gains are moderate (+2-3 F1 points), they are observed across multiple fusion strategies and visual encoders. This suggests that images contain complementary information that improved prediction when integrated appropriately.

Third, fusion design is not trivial; it plays a critical role. Particularly projected concatenation consistently outperforms simple averaging approaches. This indicates that preserving structure for each modality before integration may be more effective than naive blending of embeddings.

Fourth, caption augmentation improved performance but does not fully replace direct multimodal fusion. While captions inject visual semantics into the textual input, with the images,

we have access to visual representations that may not be fully captured by generated descriptions. Therefore, this suggests that visual features contain information that cannot be expressed linguistically.

Finally, preliminary experiments suggest that allowing limited fine-tuning of the vision encoder may increase visual contribution, indicating that fixed pretrained embeddings may be suboptimal for stance reasoning.

7.2 Limitations

Despite the improvements observed, several limitations remain.

First, performance gains are moderate relative to strong textual baselines. This suggests that multimodal integration is beneficial but not transformative in this particular dataset.

Second, the dataset size is relatively small, which constrains the effective fine-tuning of large vision encoders and may limit the model’s ability to learn complex cross-modal interactions.

Third, our experiments primarily focused on early fusion strategies. While effective, these approaches may limit deeper cross-modal reasoning compared to architectures that explicitly model cross-attention between modalities.

7.3 Future Directions

Future work could explore cross-attention multimodal architectures that enable deeper interaction between textual and visual representations.

Additionally, more advanced modality-balancing mechanisms could help mitigate textual dominance during fusion and better regulate the contribution of visual features.

Evaluating the proposed methods on larger and more diverse multimodal datasets would further clarify the generalizability of these findings.

Finally, applying more robust data augmentation strategies and improved caption generation techniques may yield additional performance improvements.

8 Conclusion

In this work, we studied whether multimodal models can improve binary stance detection in argumentative social media content. Our experiments show that strong textual baselines already achieve high performance, confirming that language carries the dominant explicit argumentative signal in the ImageArg dataset.

However, carefully designed multimodal fusion strategies yield consistent and measurable improvements over text-only models. The best multimodal configuration achieved an increase of approximately +2–3 F1 points compared to the strongest textual baseline. This demonstrates that visual cues provide complementary information that can improve stance prediction when appropriately integrated.

Our results indicate that multimodal effectiveness depends strongly on fusion design and modality balancing. Concatenation and projection strategies outperform simple averaging approaches, suggesting that preserving modality structure before integration is crucial.

Overall, multimodal modeling is beneficial in this setting, but its success depends on thoughtful architectural design rather than naive feature combination. These findings contribute to a better understanding of how textual and visual signals interact in argumentative social media analysis.

Code Availability

The full analysis pipeline (Python scripts and Jupyter notebooks) is publicly accessible at:
<https://github.com/DBlassio/multimodal-argmining>

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